CPS188: Lab 07 - Strings, Plotting and Recursion
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Problem #1:
Algorithm:
1. The user will be asked to enter a string
2. The string will be cleaned and outputted
3. The string will be reversed and outputted
4. Finally, it will be determined and outputted whether the string is a palindrome or not

Code:
```
#include <stdio.h>
#include <ctype.h>
#include <string.h>

//void function to clean the string
void clean(char previous[], char final[]) {
    int j=0;
    for (int i=0; previous[i]!='\0'; i++) {
        if(isalnum(previous[i])){
            final[j++]=tolower(previous[i]);
        }
    }
    final[j]='\0';
}

//void function to reverse the string
void reverse_helper(char previous[], char final[], int index, int
length) {
    if(index<0){
        final[length]='\0';
        return;
    }
    final[length-index-1]=previous[index];
    reverse_helper(previous, final, index-1, length);
}

//void function needed to reverse the string
void reverse(char previous[], char final[]) {
    int length = strlen(previous);
    reverse_helper(previous, final, length-1, length);
}
```

```
int main(void) {
    char string[100], cleaned[100], reversed[100]; //declaring string
variables such that each will carry max. 100 letters

    printf("User, enter a string: ");
    fgets(string, sizeof(string), stdin);
    string[strcspn(string, "\n")]='\0';

    //Calling the void function which will clean the string
    clean(string, cleaned);
    //outputting the string variable
    printf("Cleaned String: %s\n", cleaned);

    //Calling the void function which will reverse the cleaned string
    reverse(cleaned, reversed);
    //outputting the string variable
    printf("Reversed String: %s\n", reversed);

    //Check and output whether the string is or is not a palindrome
string
    if (strcmp(cleaned, reversed)==0){
        printf("The string you've entered is a palindrome.\n");
    }
    else {
        printf("This string you've enetered is not a palindrome.\n");
    }

    return (0);
}
```

Output:

```
User, enter a string: civic
Cleaned String: civic
Reversed String: civic
The string you've entered is a palindrome.

User, enter a string: You are my sunshine.
Cleaned String: youaremysunshine
Reversed String: enihsnusymerauoy
This string you've entered is not a palindrome.
```

```
User, enter a string: Too bad I hid a boot.
Cleaned String: toobadihidaboot
Reversed String: toobadihidaboot
The string you've entered is a palindrome.

User, enter a string: What is a palindrome, anyway?
Cleaned String: whatisapalindromeanyway
Reversed String: yawynaemordnilapasitahw
This string you've entered is not a palindrome.

User, enter a string: Was it a car or a cat I saw?
Cleaned String: wasitacaroracatisaw
Reversed String: wasitacaroracatisaw
The string you've entered is a palindrome.

User, enter a string: Nassau is the capital of the Bahamas
Cleaned String: nassauisthecapitalofthebahamas
Reversed String: samahabehtfolatipacehtsiuassan
This string you've entered is not a palindrome.

User, enter a string: Eva, can I stab bats in a cave?
Cleaned String: evacanistabbatsinacave
Reversed String: evacanistabbatsinacave
The string you've entered is a palindrome.
```

Screenshot:

```
User, enter a string: civic
Cleaned String: civic
Reversed String: civic
The string you've entered is a palindrome.


...Program finished with exit code 0
Press ENTER to exit console.
```

```
User, enter a string: You are my sunshine.
Cleaned String: youaremysunshine
Reversed String: enihsnusymerauoy
This string you've entered is not a palindrome.


...Program finished with exit code 0
Press ENTER to exit console.
```

```
User, enter a string: Too bad I hid a boot.
Cleaned String: toobadihidaboot
Reversed String: toobadihidaboot
The string you've entered is a palindrome.


...Program finished with exit code 0
Press ENTER to exit console.
```

```
User, enter a string: What is a palindrome, anyway?
Cleaned String: whatisapalindromeanyway
Reversed String: yawynaemordnilapasitahw
This string you've entered is not a palindrome.


...Program finished with exit code 0
Press ENTER to exit console.
```

```
User, enter a string: Was it a car or a cat I saw?
Cleaned String: wasitacaroracatisaw
Reversed String: wasitacaroracatisaw
The string you've entered is a palindrome.


...Program finished with exit code 0
Press ENTER to exit console.
```

```
User, enter a string: Nassau is the capital of the Bahamas
Cleaned String: nassauisthecapitalofthebahamas
Reversed String: samahabehtfolatipacehtsiuassan
This string you've entered is not a palindrome.


...Program finished with exit code 0
Press ENTER to exit console.
```

```
User, enter a string: Eva, can I stab bats in a cave?
Cleaned String: evacanistabbatsinacave
Reversed String: evacanistabbatsinacave
The string you've entered is a palindrome.


...Program finished with exit code 0
Press ENTER to exit console.
```

Problem #2:
Algorithm:

1. Make a C program that writes information in the text file needed to create the plot
2. Download the text file and import it into Gnuplot
3. Make a code on Gnuplot that creates a plot corresponding to the information in the data.txt file with title, axis, and legend included

Code:
C program that generates the data in the data.txt file:

```
#include <stdio.h>
#include <math.h>

//constants:
#define C 3.0e8
#define H 6.626e-34
#define K 1.38e-23

double planck_radiation(double lambda, double T){
    double lambda_m=lambda*1e-6; //converting lambda to meters
    double numerator=2*M_PI*pow(C,2)*H/pow(lambda_m,5);
    double denominator=exp((H*C)/(lambda_m*K*T))-1;
    return numerator/denominator;
}

int main(void) {
    FILE *myfile; //declaring file
    myfile=fopen("data.txt", "w"); //opening file in reading mode
    if(myfile==NULL){
```

```
            printf("Error opening file.");
            return 1;
      }

    fprintf(myfile, "# Wavelength(um)  R_3000K  R_4000K  R_5000K\n");

    for (double lambda = 0.1; lambda <= 3.0; lambda += 0.01) {
        double R_3000K = planck_radiation(lambda, 3000);
        double R_4000K = planck_radiation(lambda, 4000);
        double R_5000K = planck_radiation(lambda, 5000);
        fprintf(myfile, "%.2f  %e  %e  %e\n", lambda, R_3000K,
R_4000K, R_5000K);
    }

    fclose(myfile);//closing myfile
    printf("Data is saved to data.txt\n");
    return (0);
}
```

Gnuplot Code:

```
set terminal svg enhanced size 800,600
set title "Blackbody Radiation Spectrum"
set xlabel "Wavelength (µm)"
set ylabel "Spectral Energy Density (W/m³)"
set key top right
set xrange [0.1:3.0]
set grid

plot 'data.txt' using 1:2 with lines lc rgb "red" title "T=3,000K", \
    'data.txt' using 1:3 with lines lc rgb "blue" title "T=4,000K", \
    'data.txt' using 1:4 with lines lc rgb "green" title "T=5,000K"
```

Output:

Data generated in data.txt file from the C program:

```
# Wavelength(um)     R_3000K          R_4000K           R_5000K
0.10                 5.263136e-02     8.597098e+03
1.154028e+07
0.11                 2.570138e+00     1.409759e+05
9.832367e+07
0.12                 6.320380e+01     1.396366e+06
5.643557e+08
0.13                 9.196591e+02     9.412596e+06
2.397502e+09
```

```
0.14                 8.880579e+03       4.699908e+07
8.058946e+09
0.15                 6.188680e+04       1.849281e+08
2.250328e+10
0.16                 3.313642e+05       6.004788e+08
5.412687e+10
0.17                 1.429863e+06       1.666587e+09
1.152785e+11
0.18                 5.159746e+06       4.062554e+09
2.220760e+11
0.19                 1.603035e+07       8.885538e+09
3.934851e+11
0.20                 4.388430e+07       1.773629e+10
6.498233e+11
0.21                 1.078572e+08       3.275527e+10
1.010976e+12
0.22                 2.416481e+08       5.659496e+10
1.494635e+12
0.23                 4.997595e+08       9.232717e+10
2.114813e+12
0.24                 9.640630e+08       1.432958e+11
2.880799e+12
0.25                 1.749855e+09       2.129341e+11
3.796602e+12
0.26                 3.010523e+09       3.045676e+11
4.860878e+12
0.27                 4.940069e+09       4.212251e+11
6.067250e+12
0.28                 7.772979e+09       5.654740e+11
7.404926e+12
0.29                 1.178124e+10       7.392919e+11
8.859499e+12
0.30                 1.726863e+10       9.439802e+11
1.041384e+13
0.31                 2.456266e+10       1.180121e+12
1.204901e+13
0.32                 3.400478e+10       1.447576e+12
1.374507e+13
0.33                 4.593951e+10       1.745515e+12
1.548188e+13
```

```
0.34            6.070327e+10    2.072476e+12
1.723969e+13
0.35            7.861350e+10    2.426446e+12
1.899971e+13
0.36            9.995869e+10    2.804948e+12
2.074444e+13
0.37            1.249898e+11    3.205141e+12
2.245798e+13
0.38            1.539132e+11    3.623916e+12
2.412618e+13
0.39            1.868857e+11    4.057987e+12
2.573676e+13
0.40            2.240109e+11    4.503984e+12
2.727929e+13
0.41            2.653382e+11    4.958523e+12
2.874520e+13
0.42            3.108622e+11    5.418279e+12
3.012763e+13
0.43            3.605251e+11    5.880038e+12
3.142138e+13
0.44            4.142189e+11    6.340745e+12
3.262274e+13
0.45            4.717894e+11    6.797539e+12
3.372936e+13
0.46            5.330410e+11    7.247781e+12
3.474010e+13
0.47            5.977414e+11    7.689066e+12
3.565489e+13
0.48            6.656274e+11    8.119238e+12
3.647454e+13
0.49            7.364101e+11    8.536394e+12
3.720069e+13
0.50            8.097804e+11    8.938879e+12
3.783559e+13
0.51            8.854147e+11    9.325282e+12
3.838205e+13
0.52            9.629794e+11    9.694428e+12
3.884329e+13
0.53            1.042136e+12    1.004536e+13
3.922285e+13
```

```
0.54                1.122545e+12       1.037734e+13
3.952454e+13
0.55                1.203871e+12       1.068982e+13
3.975234e+13
0.56                1.285783e+12       1.098242e+13
3.991030e+13
0.57                1.367962e+12       1.125494e+13
4.000257e+13
0.58                1.450099e+12       1.150730e+13
4.003326e+13
0.59                1.531901e+12       1.173959e+13
4.000647e+13
0.60                1.613088e+12       1.195198e+13
3.992623e+13
0.61                1.693399e+12       1.214475e+13
3.979647e+13
0.62                1.772591e+12       1.231828e+13
3.962099e+13
0.63                1.850438e+12       1.247301e+13
3.940348e+13
0.64                1.926733e+12       1.260946e+13
3.914747e+13
0.65                2.001290e+12       1.272817e+13
3.885634e+13
0.66                2.073940e+12       1.282975e+13
3.853331e+13
0.67                2.144531e+12       1.291484e+13
3.818144e+13
0.68                2.212933e+12       1.298409e+13
3.780363e+13
0.69                2.279030e+12       1.303819e+13
3.740259e+13
0.70                2.342725e+12       1.307781e+13
3.698090e+13
0.71                2.403934e+12       1.310365e+13
3.654097e+13
0.72                2.462592e+12       1.311641e+13
3.608504e+13
0.73                2.518646e+12       1.311677e+13
3.561523e+13
```

```
0.74                2.572056e+12       1.310542e+13
3.513349e+13
0.75                2.622795e+12       1.308303e+13
3.464164e+13
0.76                2.670849e+12       1.305025e+13
3.414137e+13
0.77                2.716213e+12       1.300772e+13
3.363424e+13
0.78                2.758892e+12       1.295608e+13
3.312171e+13
0.79                2.798900e+12       1.289593e+13
3.260508e+13
0.80                2.836260e+12       1.282784e+13
3.208560e+13
0.81                2.871002e+12       1.275239e+13
3.156437e+13
0.82                2.903161e+12       1.267012e+13
3.104243e+13
0.83                2.932780e+12       1.258155e+13
3.052071e+13
0.84                2.959905e+12       1.248719e+13
3.000006e+13
0.85                2.984588e+12       1.238749e+13
2.948127e+13
0.86                3.006886e+12       1.228293e+13
2.896503e+13
0.87                3.026856e+12       1.217393e+13
2.845198e+13
0.88                3.044560e+12       1.206091e+13
2.794268e+13
0.89                3.060062e+12       1.194426e+13
2.743766e+13
0.90                3.073429e+12       1.182435e+13
2.693737e+13
0.91                3.084726e+12       1.170152e+13
2.644222e+13
0.92                3.094022e+12       1.157611e+13
2.595256e+13
0.93                3.101387e+12       1.144844e+13
2.546872e+13
```

```
0.94                 3.106890e+12      1.131879e+13
2.499096e+13
0.95                 3.110600e+12      1.118744e+13
2.451954e+13
0.96                 3.112586e+12      1.105466e+13
2.405466e+13
0.97                 3.112917e+12      1.092068e+13
2.359650e+13
0.98                 3.111662e+12      1.078574e+13
2.314519e+13
0.99                 3.108889e+12      1.065005e+13
2.270086e+13
1.00                 3.104664e+12      1.051381e+13
2.226361e+13
1.01                 3.099052e+12      1.037720e+13
2.183351e+13
1.02                 3.092119e+12      1.024040e+13
2.141062e+13
1.03                 3.083928e+12      1.010357e+13
2.099497e+13
1.04                 3.074539e+12      9.966867e+12
2.058658e+13
1.05                 3.064014e+12      9.830420e+12
2.018545e+13
1.06                 3.052412e+12      9.694362e+12
1.979159e+13
1.07                 3.039790e+12      9.558811e+12
1.940495e+13
1.08                 3.026203e+12      9.423877e+12
1.902552e+13
1.09                 3.011705e+12      9.289662e+12
1.865325e+13
1.10                 2.996350e+12      9.156259e+12
1.828808e+13
1.11                 2.980188e+12      9.023751e+12
1.792996e+13
1.12                 2.963268e+12      8.892217e+12
1.757882e+13
1.13                 2.945637e+12      8.761728e+12
1.723459e+13
```

1.14	2.927341e+12	8.632348e+12	1.689719e+13
1.15	2.908425e+12	8.504134e+12	1.656653e+13
1.16	2.888932e+12	8.377141e+12	1.624253e+13
1.17	2.868901e+12	8.251414e+12	1.592510e+13
1.18	2.848373e+12	8.126996e+12	1.561413e+13
1.19	2.827384e+12	8.003924e+12	1.530954e+13
1.20	2.805973e+12	7.882232e+12	1.501122e+13
1.21	2.784173e+12	7.761948e+12	1.471908e+13
1.22	2.762018e+12	7.643096e+12	1.443300e+13
1.23	2.739539e+12	7.525700e+12	1.415289e+13
1.24	2.716768e+12	7.409776e+12	1.387864e+13
1.25	2.693734e+12	7.295340e+12	1.361015e+13
1.26	2.670464e+12	7.182404e+12	1.334731e+13
1.27	2.646985e+12	7.070977e+12	1.309001e+13
1.28	2.623323e+12	6.961067e+12	1.283815e+13
1.29	2.599501e+12	6.852677e+12	1.259162e+13
1.30	2.575544e+12	6.745810e+12	1.235032e+13
1.31	2.551472e+12	6.640467e+12	1.211415e+13
1.32	2.527307e+12	6.536647e+12	1.188300e+13
1.33	2.503068e+12	6.434346e+12	1.165678e+13

```
1.34                 2.478775e+12       6.333560e+12
1.143537e+13
1.35                 2.454445e+12       6.234282e+12
1.121868e+13
1.36                 2.430095e+12       6.136506e+12
1.100661e+13
1.37                 2.405742e+12       6.040223e+12
1.079907e+13
1.38                 2.381399e+12       5.945423e+12
1.059595e+13
1.39                 2.357082e+12       5.852096e+12
1.039717e+13
1.40                 2.332805e+12       5.760229e+12
1.020262e+13
1.41                 2.308579e+12       5.669811e+12
1.001222e+13
1.42                 2.284417e+12       5.580829e+12
9.825882e+12
1.43                 2.260331e+12       5.493268e+12
9.643509e+12
1.44                 2.236330e+12       5.407114e+12
9.465016e+12
1.45                 2.212426e+12       5.322353e+12
9.290319e+12
1.46                 2.188626e+12       5.238969e+12
9.119333e+12
1.47                 2.164941e+12       5.156945e+12
8.951975e+12
1.48                 2.141378e+12       5.076267e+12
8.788164e+12
1.49                 2.117946e+12       4.996917e+12
8.627822e+12
1.50                 2.094650e+12       4.918878e+12
8.470870e+12
1.51                 2.071498e+12       4.842134e+12
8.317232e+12
1.52                 2.048497e+12       4.766667e+12
8.166833e+12
1.53                 2.025651e+12       4.692459e+12
8.019601e+12
```

```
1.54                 2.002966e+12       4.619494e+12
7.875463e+12
1.55                 1.980447e+12       4.547754e+12
7.734349e+12
1.56                 1.958099e+12       4.477221e+12
7.596192e+12
1.57                 1.935926e+12       4.407877e+12
7.460923e+12
1.58                 1.913931e+12       4.339704e+12
7.328478e+12
1.59                 1.892118e+12       4.272686e+12
7.198792e+12
1.60                 1.870491e+12       4.206804e+12
7.071802e+12
1.61                 1.849051e+12       4.142040e+12
6.947447e+12
1.62                 1.827802e+12       4.078378e+12
6.825667e+12
1.63                 1.806745e+12       4.015799e+12
6.706403e+12
1.64                 1.785884e+12       3.954286e+12
6.589599e+12
1.65                 1.765219e+12       3.893823e+12
6.475197e+12
1.66                 1.744752e+12       3.834391e+12
6.363144e+12
1.67                 1.724485e+12       3.775975e+12
6.253386e+12
1.68                 1.704418e+12       3.718556e+12
6.145871e+12
1.69                 1.684553e+12       3.662119e+12
6.040548e+12
1.70                 1.664890e+12       3.606647e+12
5.937368e+12
1.71                 1.645430e+12       3.552123e+12
5.836280e+12
1.72                 1.626172e+12       3.498532e+12
5.737239e+12
1.73                 1.607118e+12       3.445858e+12
5.640198e+12
```

```
1.74                 1.588268e+12       3.394084e+12
5.545111e+12
1.75                 1.569621e+12       3.343195e+12
5.451935e+12
1.76                 1.551177e+12       3.293176e+12
5.360626e+12
1.77                 1.532936e+12       3.244011e+12
5.271142e+12
1.78                 1.514897e+12       3.195687e+12
5.183442e+12
1.79                 1.497060e+12       3.148186e+12
5.097486e+12
1.80                 1.479424e+12       3.101496e+12
5.013235e+12
1.81                 1.461989e+12       3.055602e+12
4.930650e+12
1.82                 1.444753e+12       3.010490e+12
4.849695e+12
1.83                 1.427716e+12       2.966146e+12
4.770333e+12
1.84                 1.410877e+12       2.922555e+12
4.692528e+12
1.85                 1.394234e+12       2.879706e+12
4.616245e+12
1.86                 1.377786e+12       2.837583e+12
4.541451e+12
1.87                 1.361533e+12       2.796175e+12
4.468113e+12
1.88                 1.345473e+12       2.755468e+12
4.396199e+12
1.89                 1.329604e+12       2.715450e+12
4.325676e+12
1.90                 1.313926e+12       2.676108e+12
4.256515e+12
1.91                 1.298436e+12       2.637430e+12
4.188685e+12
1.92                 1.283134e+12       2.599404e+12
4.122157e+12
1.93                 1.268017e+12       2.562018e+12
4.056902e+12
```

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1.94                  1.253085e+12       2.525261e+12
3.992893e+12
1.95                  1.238336e+12       2.489120e+12
3.930102e+12
1.96                  1.223767e+12       2.453586e+12
3.868503e+12
1.97                  1.209379e+12       2.418646e+12
3.808069e+12
1.98                  1.195168e+12       2.384290e+12
3.748776e+12
1.99                  1.181133e+12       2.350508e+12
3.690599e+12
2.00                  1.167273e+12       2.317288e+12
3.633513e+12
2.01                  1.153585e+12       2.284621e+12
3.577495e+12
2.02                  1.140069e+12       2.252496e+12
3.522522e+12
2.03                  1.126722e+12       2.220904e+12
3.468572e+12
2.04                  1.113543e+12       2.189835e+12
3.415622e+12
2.05                  1.100529e+12       2.159280e+12
3.363652e+12
2.06                  1.087679e+12       2.129228e+12
3.312640e+12
2.07                  1.074992e+12       2.099671e+12
3.262566e+12
2.08                  1.062465e+12       2.070600e+12
3.213410e+12
2.09                  1.050097e+12       2.042005e+12
3.165153e+12
2.10                  1.037887e+12       2.013879e+12
3.117775e+12
2.11                  1.025831e+12       1.986213e+12
3.071259e+12
2.12                  1.013928e+12       1.958998e+12
3.025585e+12
2.13                  1.002178e+12       1.932225e+12
2.980736e+12
```

2.14	9.905770e+11	1.905888e+12	2.936696e+12
2.15	9.791244e+11	1.879979e+12	2.893446e+12
2.16	9.678182e+11	1.854488e+12	2.850971e+12
2.17	9.566566e+11	1.829409e+12	2.809255e+12
2.18	9.456379e+11	1.804735e+12	2.768282e+12
2.19	9.347605e+11	1.780457e+12	2.728036e+12
2.20	9.240224e+11	1.756570e+12	2.688502e+12
2.21	9.134222e+11	1.733065e+12	2.649666e+12
2.22	9.029579e+11	1.709936e+12	2.611514e+12
2.23	8.926280e+11	1.687176e+12	2.574031e+12
2.24	8.824308e+11	1.664778e+12	2.537204e+12
2.25	8.723646e+11	1.642737e+12	2.501019e+12
2.26	8.624278e+11	1.621045e+12	2.465464e+12
2.27	8.526186e+11	1.599696e+12	2.430526e+12
2.28	8.429355e+11	1.578684e+12	2.396191e+12
2.29	8.333769e+11	1.558003e+12	2.362449e+12
2.30	8.239412e+11	1.537647e+12	2.329286e+12
2.31	8.146267e+11	1.517611e+12	2.296693e+12
2.32	8.054320e+11	1.497888e+12	2.264656e+12
2.33	7.963554e+11	1.478474e+12	2.233165e+12

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2.34                 7.873954e+11        1.459361e+12
2.202209e+12
2.35                 7.785505e+11        1.440546e+12
2.171778e+12
2.36                 7.698191e+11        1.422023e+12
2.141861e+12
2.37                 7.611998e+11        1.403787e+12
2.112448e+12
2.38                 7.526911e+11        1.385832e+12
2.083529e+12
2.39                 7.442916e+11        1.368154e+12
2.055094e+12
2.40                 7.359997e+11        1.350748e+12
2.027135e+12
2.41                 7.278140e+11        1.333609e+12
1.999640e+12
2.42                 7.197331e+11        1.316733e+12
1.972602e+12
2.43                 7.117557e+11        1.300114e+12
1.946012e+12
2.44                 7.038803e+11        1.283748e+12
1.919861e+12
2.45                 6.961055e+11        1.267631e+12
1.894140e+12
2.46                 6.884301e+11        1.251758e+12
1.868841e+12
2.47                 6.808526e+11        1.236126e+12
1.843957e+12
2.48                 6.733718e+11        1.220730e+12
1.819478e+12
2.49                 6.659863e+11        1.205566e+12
1.795398e+12
2.50                 6.586948e+11        1.190629e+12
1.771708e+12
2.51                 6.514962e+11        1.175917e+12
1.748402e+12
2.52                 6.443890e+11        1.161425e+12
1.725472e+12
2.53                 6.373721e+11        1.147149e+12
1.702910e+12
```
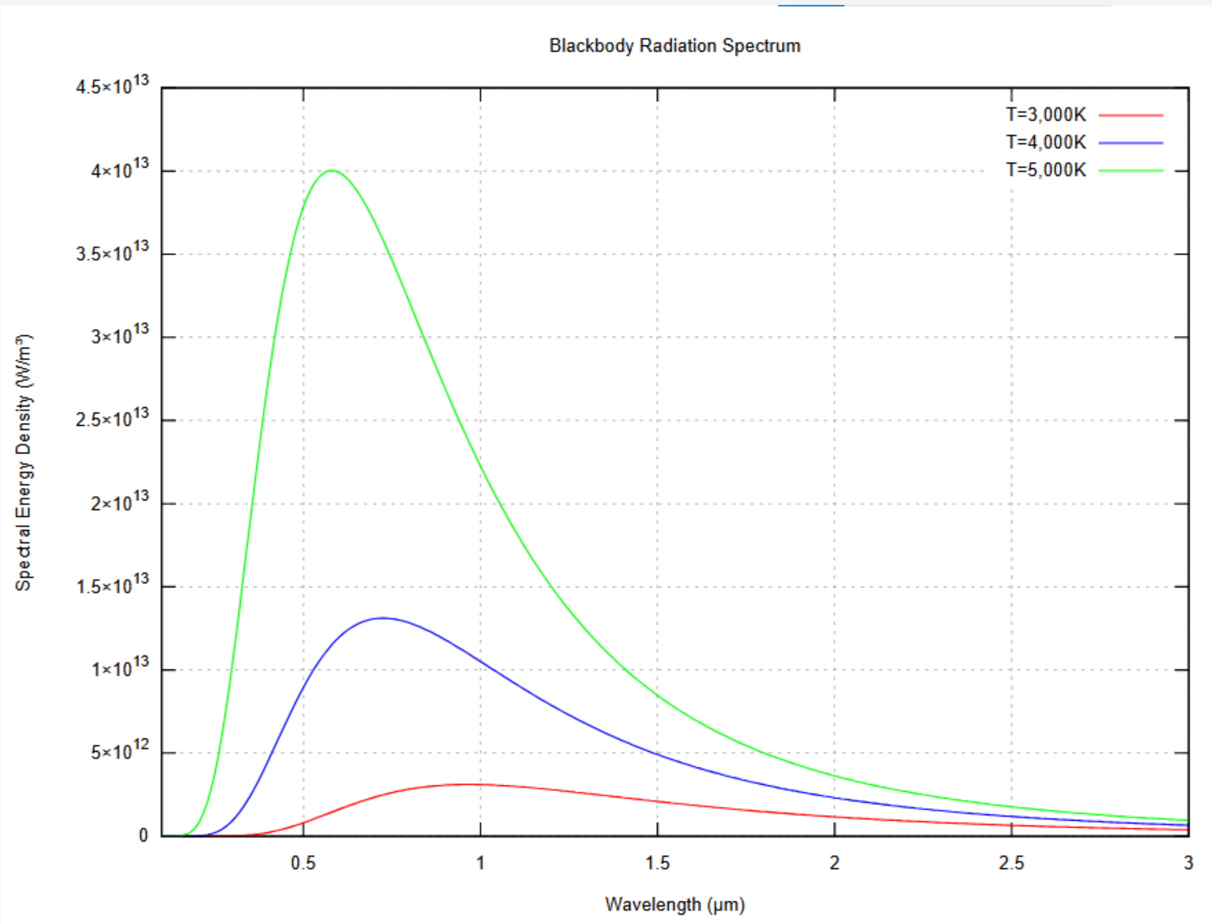

```
2.54                 6.304442e+11      1.133086e+12
1.680711e+12
2.55                 6.236041e+11      1.119231e+12
1.658866e+12
2.56                 6.168507e+11      1.105582e+12
1.637370e+12
2.57                 6.101827e+11      1.092135e+12
1.616216e+12
2.58                 6.035990e+11      1.078886e+12
1.595398e+12
2.59                 5.970985e+11      1.065832e+12
1.574909e+12
2.60                 5.906799e+11      1.052970e+12
1.554742e+12
2.61                 5.843422e+11      1.040297e+12
1.534893e+12
2.62                 5.780842e+11      1.027808e+12
1.515356e+12
2.63                 5.719049e+11      1.015502e+12
1.496123e+12
2.64                 5.658032e+11      1.003375e+12
1.477191e+12
2.65                 5.597780e+11      9.914236e+11
1.458553e+12
2.66                 5.538284e+11      9.796455e+11
1.440204e+12
2.67                 5.479531e+11      9.680375e+11
1.422139e+12
2.68                 5.421513e+11      9.565969e+11
1.404353e+12
2.69                 5.364218e+11      9.453207e+11
1.386840e+12
2.70                 5.307638e+11      9.342063e+11
1.369595e+12
2.71                 5.251763e+11      9.232510e+11
1.352615e+12
2.72                 5.196582e+11      9.124521e+11
1.335893e+12
2.73                 5.142086e+11      9.018071e+11
1.319426e+12
```

```
2.74                 5.088267e+11       8.913133e+11
1.303208e+12
2.75                 5.035113e+11       8.809684e+11
1.287236e+12
2.76                 4.982617e+11       8.707697e+11
1.271504e+12
2.77                 4.930770e+11       8.607150e+11
1.256009e+12
2.78                 4.879562e+11       8.508019e+11
1.240747e+12
2.79                 4.828985e+11       8.410280e+11
1.225713e+12
2.80                 4.779030e+11       8.313910e+11
1.210903e+12
2.81                 4.729688e+11       8.218888e+11
1.196314e+12
2.82                 4.680952e+11       8.125191e+11
1.181941e+12
2.83                 4.632812e+11       8.032798e+11
1.167781e+12
2.84                 4.585262e+11       7.941688e+11
1.153830e+12
2.85                 4.538292e+11       7.851839e+11
1.140084e+12
2.86                 4.491894e+11       7.763232e+11
1.126540e+12
2.87                 4.446062e+11       7.675847e+11
1.113194e+12
2.88                 4.400787e+11       7.589663e+11
1.100043e+12
2.89                 4.356062e+11       7.504662e+11
1.087083e+12
2.90                 4.311878e+11       7.420824e+11
1.074312e+12
2.91                 4.268230e+11       7.338132e+11
1.061726e+12
2.92                 4.225109e+11       7.256566e+11
1.049321e+12
2.93                 4.182508e+11       7.176109e+11
1.037095e+12
```

```
2.94            4.140421e+11        7.096743e+11
1.025045e+12
2.95            4.098840e+11        7.018451e+11
1.013167e+12
2.96            4.057758e+11        6.941216e+11
1.001460e+12
2.97            4.017169e+11        6.865021e+11
9.899188e+11
2.98            3.977066e+11        6.789850e+11
9.785419e+11
2.99            3.937442e+11        6.715686e+11
9.673264e+11
3.00            3.898291e+11        6.642514e+11
9.562695e+11
```

Gnuplot output (screenshot):



Screenshot:

hello.c data.txt

```
# Wavelength(um)        R_3000K             R_4000K             R_5000K
0.10                    5.263136e-02        8.597098e+03        1.154028e+07
0.11                    2.570138e+00        1.409759e+05        9.832367e+07
0.12                    6.320380e+01        1.396366e+06        5.643557e+08
0.13                    9.196591e+02        9.412596e+06        2.397502e+09
0.14                    8.880579e+03        4.699908e+07        8.058946e+09
0.15                    6.188680e+04        1.849281e+08        2.250328e+10
0.16                    3.313642e+05        6.004788e+08        5.412687e+10
0.17                    1.429863e+06        1.666587e+09        1.152785e+11
0.18                    5.159746e+06        4.062554e+09        2.220760e+11
0.19                    1.603035e+07        8.885538e+09        3.934851e+11
0.20                    4.388430e+07        1.773629e+10        6.498233e+11
0.21                    1.078572e+08        3.275527e+10        1.010976e+12
0.22                    2.416481e+08        5.659496e+10        1.494635e+12
0.23                    4.997595e+08        9.232717e+10        2.114813e+12
0.24                    9.640630e+08        1.432958e+11        2.880799e+12
0.25                    1.749855e+09        2.129341e+11        3.796602e+12
0.26                    3.010523e+09        3.045676e+11        4.860878e+12
0.27                    4.940069e+09        4.212251e+11        6.067250e+12
0.28                    7.772979e+09        5.654740e+11        7.404926e+12
0.29                    1.178124e+10        7.392919e+11        8.859499e+12
0.30                    1.726863e+10        9.439802e+11        1.041384e+13
0.31                    2.456266e+10        1.180121e+12        1.204901e+13
0.32                    3.400478e+10        1.447576e+12        1.374507e+13
0.33                    4.593951e+10        1.745515e+12        1.548188e+13
0.34                    6.070327e+10        2.072476e+12        1.723969e+13
0.35                    7.861350e+10        2.426446e+12        1.899971e+13
0.36                    9.995869e+10        2.804948e+12        2.074444e+13
0.37                    1.249898e+11        3.205141e+12        2.245798e+13
0.38                    1.539132e+11        3.623916e+12        2.412618e+13
0.39                    1.868857e+11        4.057987e+12        2.573676e+13
```

```
-o "hello" "hello.c" (in directory: C:\Users\User\Desktop\hello)
n finished successfully.
```